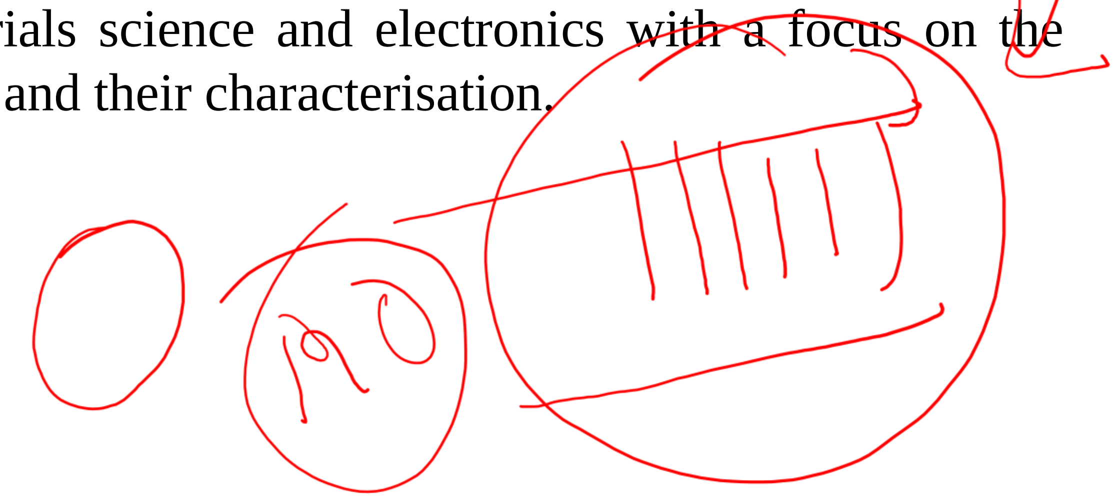
Solid State Chemistry

Solid state

- **Definite volume and shape**
- **Strong intermolecular forces**
- **Low kinetic energy**

Solid state compound

- **Solid-state chemistry** is the study of the synthesis, structure, and properties of solid phase materials, particularly, but not necessarily exclusively of, ~~non-~~molecular solids.
- It has a strong overlap with solid state physics, mineralogy, crystallography, ceramics, metallurgy, materials science and electronics with a focus on the synthesis of novel materials and their characterisation.



- **Materials chemistry**

Classification of Solids

- Solids can be classified as crystalline or amorphous.

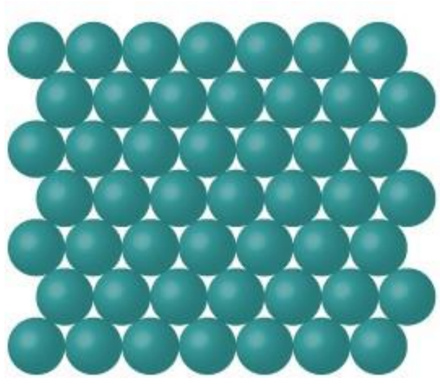
(on the basis of arrangement of constituent atoms/molecules/ions)

- A crystalline solid is composed of one or more crystals; each crystal has a well-defined, ordered structure in three dimensions. They have sharp melting point

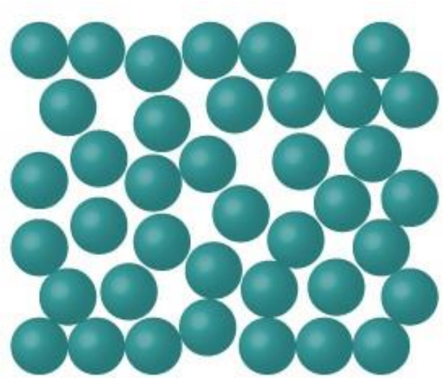
Examples include sodium chloride and sucrose.

- An amorphous solid has a disordered structure. It lacks the well-defined arrangement of basic units found in a crystal. They melt over a wide range of temperature.

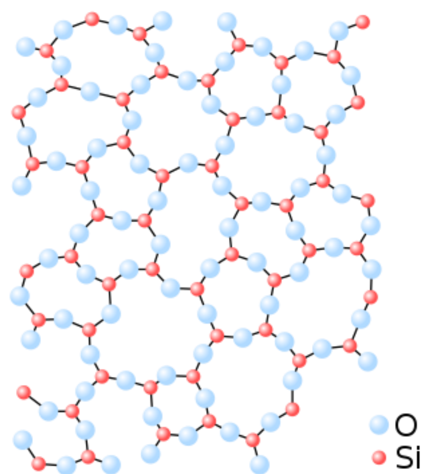
- Glass is an example of amorphous solid.



Crystalline



Amorphous



Classification of Solids

Solids can be classified into following types (based on nature of bonding)

- **Molecular solids (Van der Waals forces)**
- **Covalent (Covalent bond)**
- **Metallic (Metallic bond)**
- **Ionic (Ionic bond)**

Molecular solids

- **A molecular solid is a solid that consists of atoms or molecules held together by weak intermolecular forces.**

- **Soft, low melting point, volatile, electrical insulators.**

- **Solid neon (melting point of -248^{o}C), Solid CO_2 (dry ice), Solid H_2O (ice), Iodine, Sugar**

- **In non-polar molecules, the inter- molecular forces are only weak van der Waals forces.**

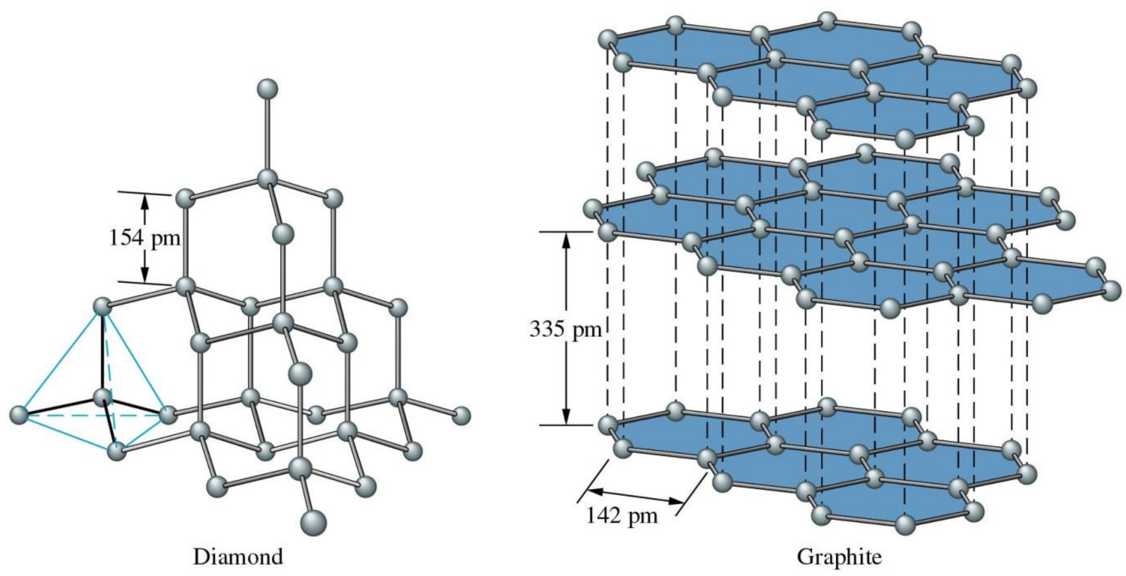
- **Polar molecules have dipoles and so have slightly stronger attractions between them meaning that the melting point of a polar molecular solid is a little higher because more energy is needed to overcome the slightly stronger forces.**

Covalent Solids

- A **covalent network solid** is a solid that consists of atoms held together in large networks or chains by covalent bonds.

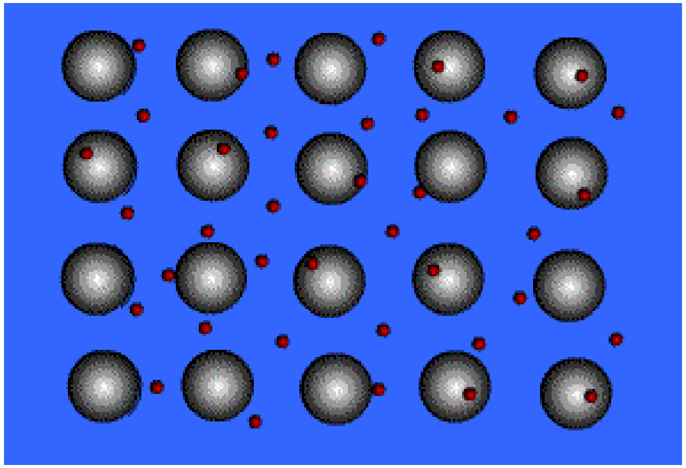
- Very hard, high melting point, poor conductors of heat and electricity.

- Examples include carbon, in its forms as diamond or graphite, asbestos, and silicon carbide.



Metallic Solids

- A metallic solid is a solid that consists of positive metal ions held together by a surrounding "sea of electrons "(metallic bonding).



- In this kind of bonding, positively charged metal ions form the lattice which is submerged in the "sea of electrons"

Or, the metal ions occupy fixed position in the lattice structure and they are surrounded by delocalized electrons.

- Very soft to very hard, low to high melting point, good conductors of electricity and heat.
- Malleable and ductile due to non-directional metallic bonds
- Good conductors of heat and electricity due to high density and mobile electrons
- High melting and boiling points due to strong metallic bonds

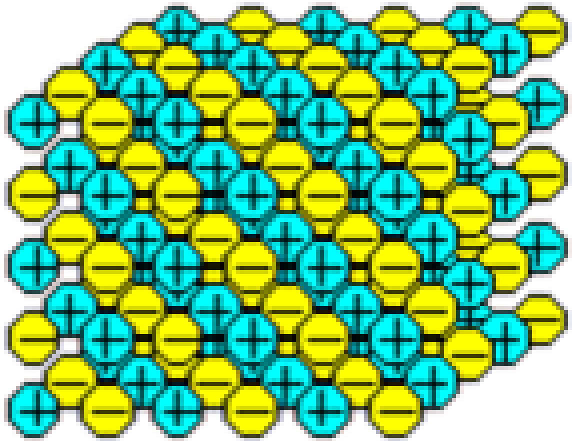
iron, copper, and silver

Ionic Solids

An ionic solid is a solid that consists of cations and anions held together by electrostatic attraction of opposite charges (ionic bond).

•Brittle, high melting point, good conductors in the aqueous solution or fused state, high heats of fusion.

Salts like NaCl, KNO_3, LiF, $BaSO_4$ etc.



Ionic lattice structure

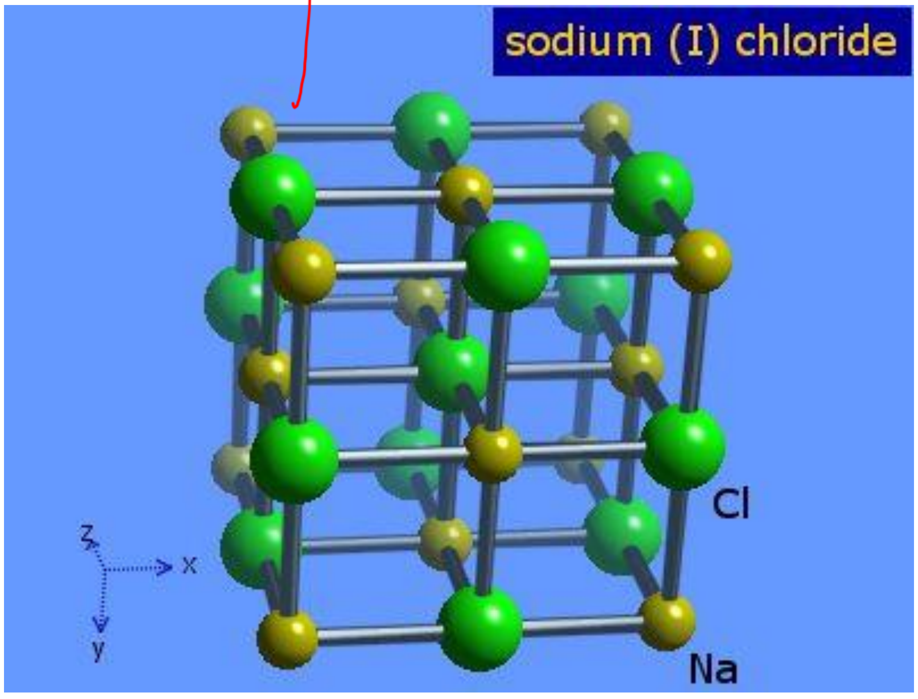
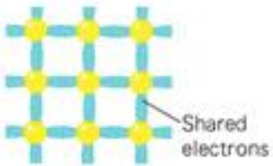
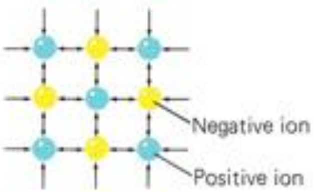
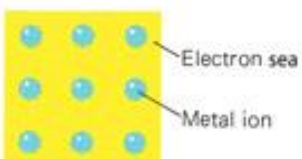
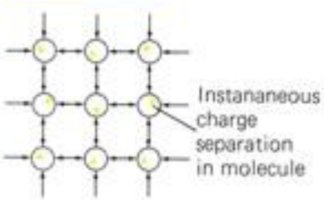


Table 10-1 Crystal Types (Bonds Are Strongest in Covalent Crystals, Weakest in Molecular Crystals)

Type		Bond	Example	Properties
Covalent		Shared electrons	Diamond C	Very hard; high melting point
Ionic		Electrical attraction	Sodium chloride NaCl	Hard; high melting point
Metallic		Electron sea	Copper Cu	Can be deformed; metallic luster; high electrical and thermal conductivity
Molecular		Van der Waals forces	Ice H_2O	Soft; low melting point

Structure ?

Crystal Lattices and Unit cell

➢The idea of lattice was developed from internal regularity of a crystal structure.

➢A highly ordered three dimensional structure formed by atoms ions or molecules are called crystal lattice. If the constituents are denoted by points, then a lattice may be regarded as an infinite set of points repeated at a regularly through space.

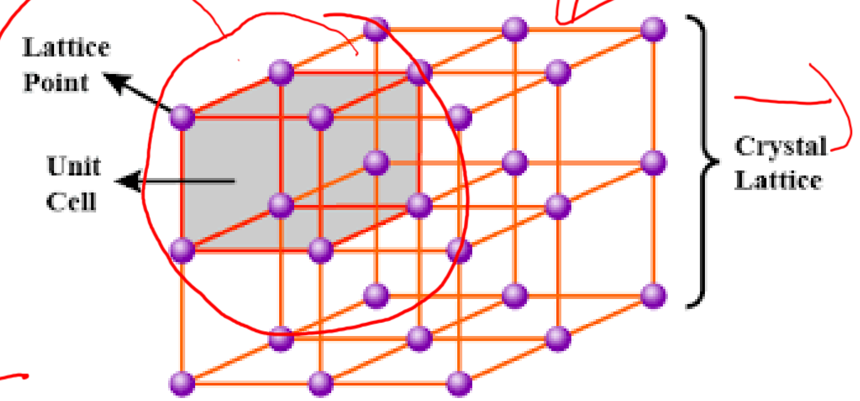
crystal structure

➢1-D Lattice: along lines
➢2D Lattice: along a plane
➢3D Lattice: along three coordinate axis

➢ Unit cell is the smallest building unit of a crystal, which when stacked together with pure translational repetition reproduces the whole crystal. It is an essential feature of crystal structure.

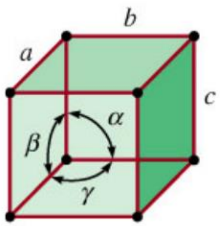
➢Any point placed in unit cell must occupy
The same relative position in every unit cell

Unit cell

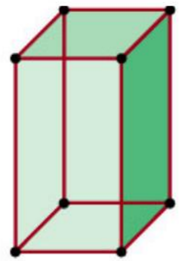


The 7 Crystal Systems

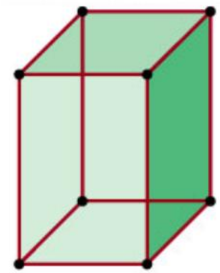
All 3D crystals belong to one of 7 *crystal systems*.



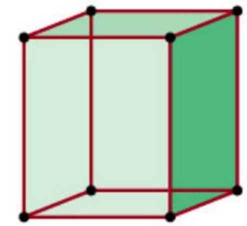
Cubic
$a = b = c$
$\alpha = \beta = \gamma = 90°$



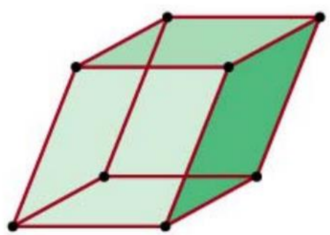
Tetragonal
$a = b \neq c$
$\alpha = \beta = \gamma = 90°$



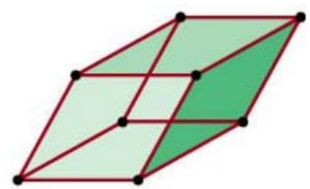
Orthorhombic
$a \neq b \neq c$
$\alpha = \beta = \gamma = 90°$



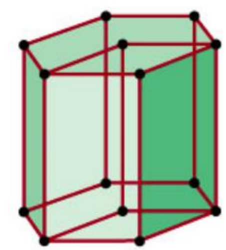
Monoclinic
$a \neq b \neq c$
$\gamma \neq \alpha = \beta = 90°$



Triclinic
$a \neq b \neq c$
$\alpha \neq \beta \neq \gamma \neq 90°$



Rhombohedral
$a = b = c$
$\alpha = \beta = \gamma \neq 90°$

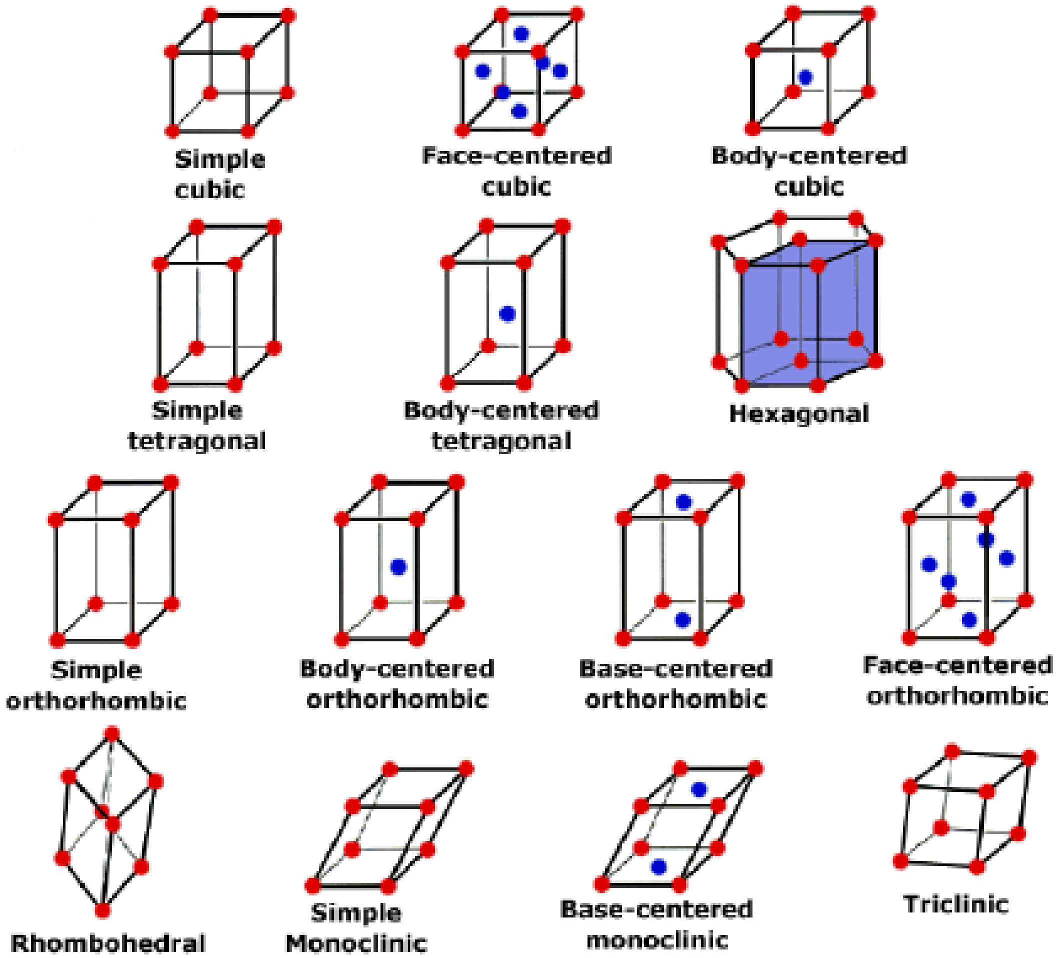


Hexagonal
$a = b \neq c$
$\alpha = \beta = 90°,\ \gamma = 120°$

Bravais Lattices

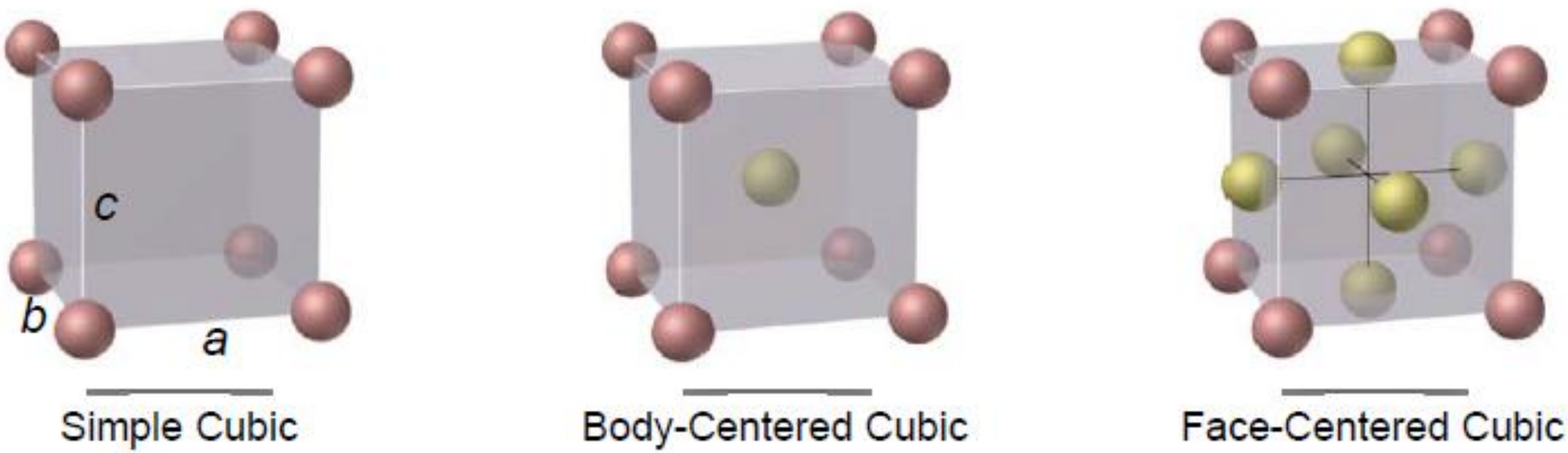
All 3D crystals belong to one of 14 Bravais lattices.

***Bravais lattice*: An infinite array of points with an arrangement and orientation that looks exactly the same from any lattice point.**



Simple (P)
Bodycentered (I)
Face centered (F)

Cubic Lattices

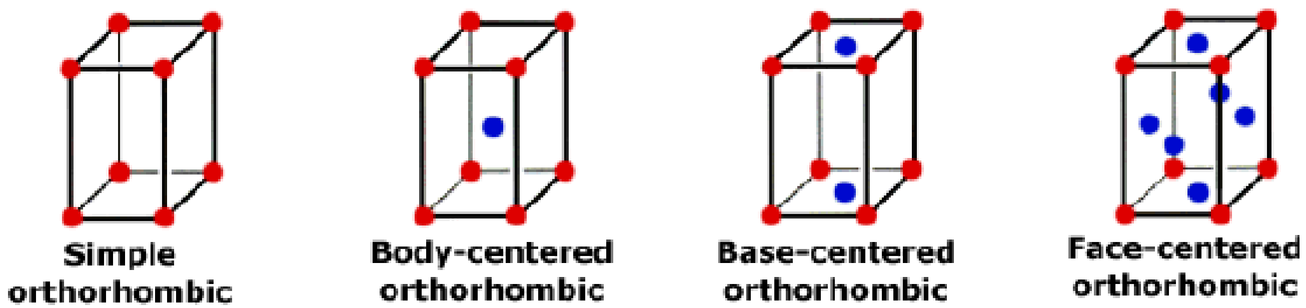


- The lengths of the unit cell edges (*a*,*b*,*c*) are called *lattice constants*.
- For cubic crystals, $a = b = c$, so there is only one lattice constant (a).

Contents of Unit cell in orthorhombic

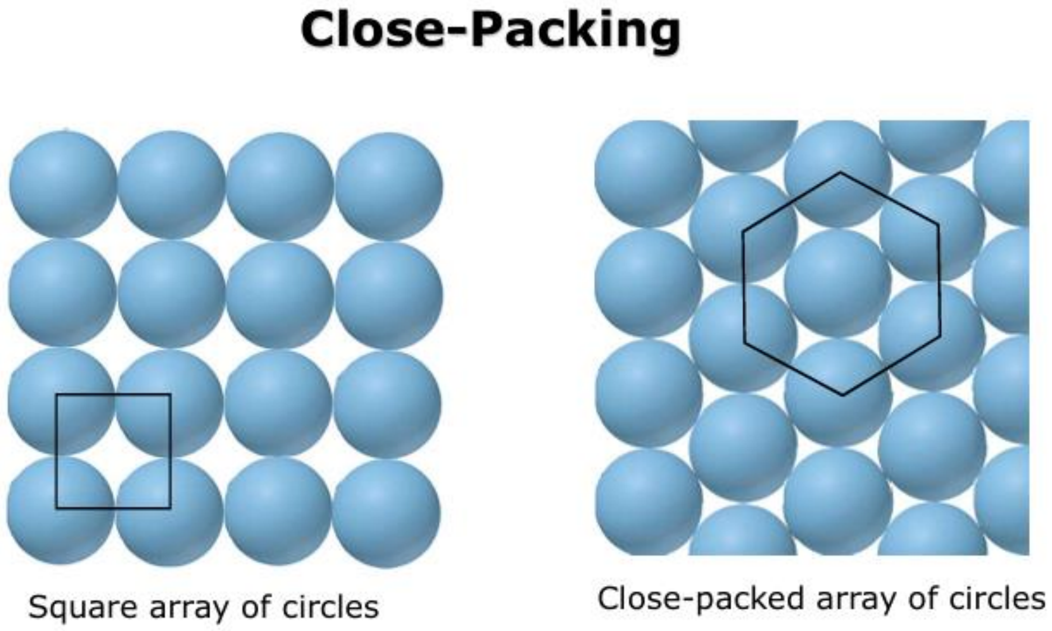
An important feature of the unit cell is the number of lattice points it contains. Atoms/ions/molecules are often located at lattice points.

Atoms	Shared Between:	Each atom counts:
corner	8 cells	1/8
face center	2 cells	1/2
body center	1 cell	1
edge center	4 cells	1/4

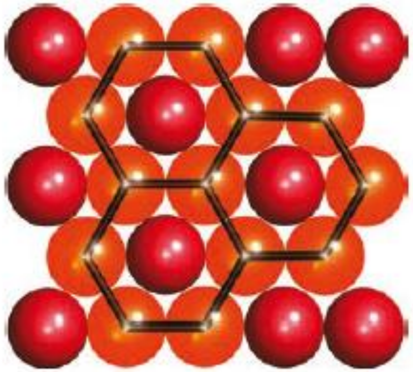


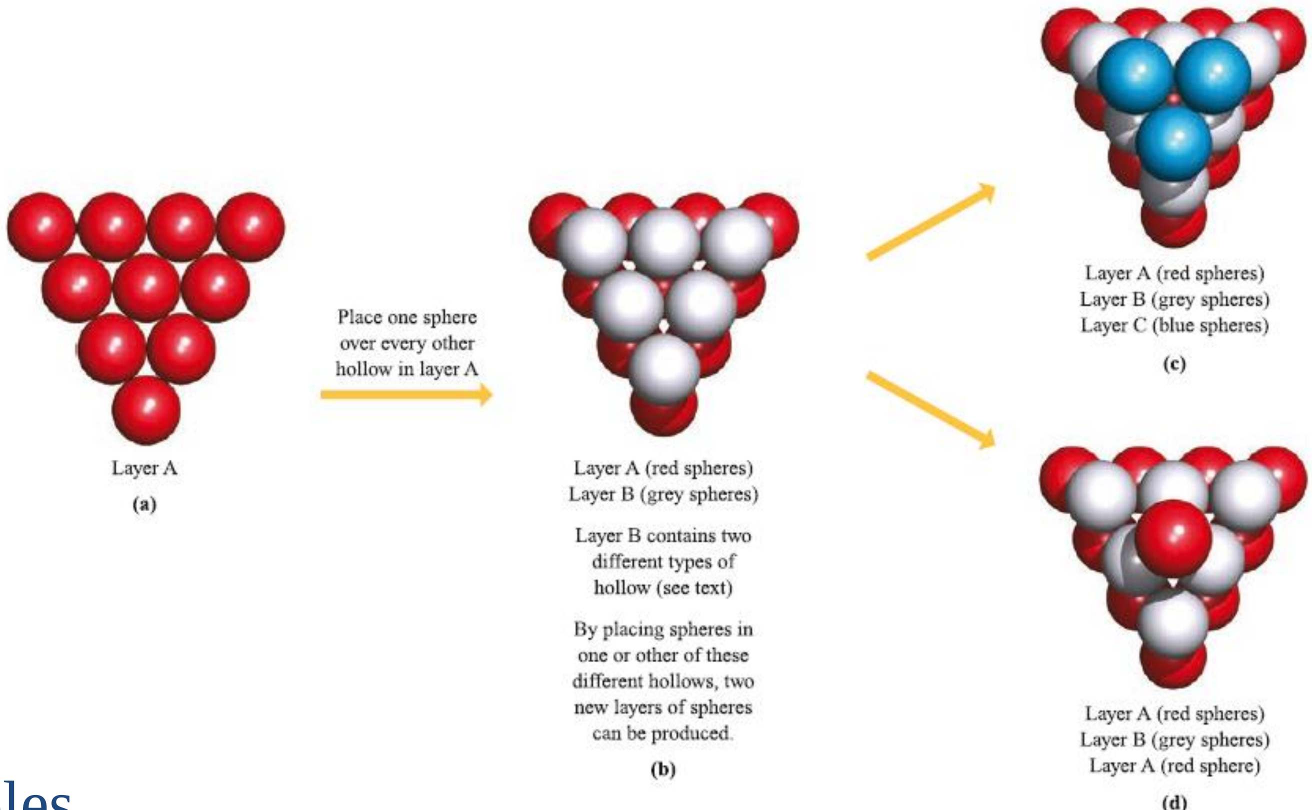
Simple orthorhombic	Body-centered orthorhombic	Base-centered orthorhombic	Face-centered orthorhombic
$8 \times 1/8 = \mathbf{1}$	$8 \times 1/8 + 1 \times 1 = \mathbf{2}$	$8 \times 1/8 + 2 \times 1/2 = \mathbf{2}$	$8 \times 1/8 + 4 \times 1/2 = \mathbf{4}$

Packing of Spheres

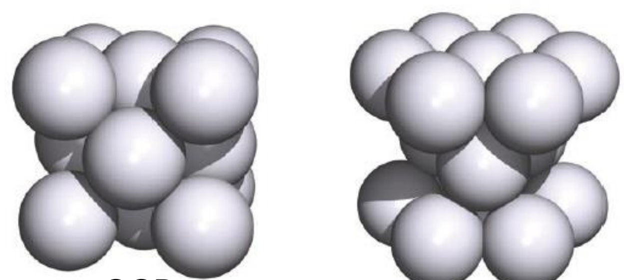


Packing of Spheres

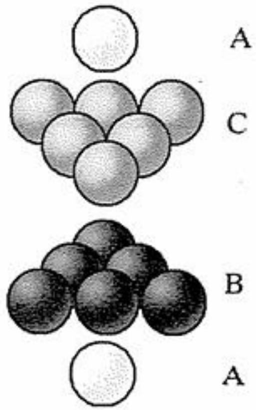


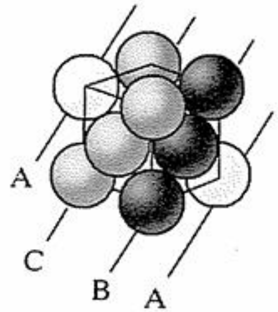


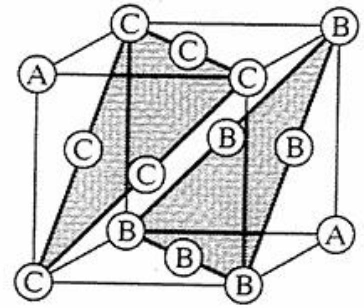
- Octahedral holes
- Tetrahedral holes



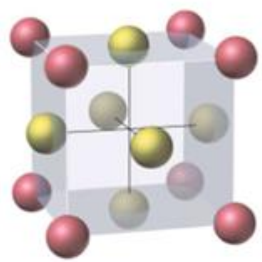
CCP HCP



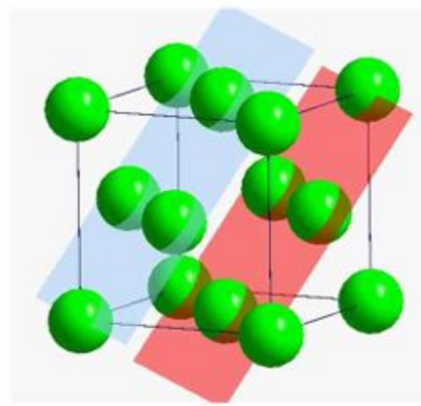




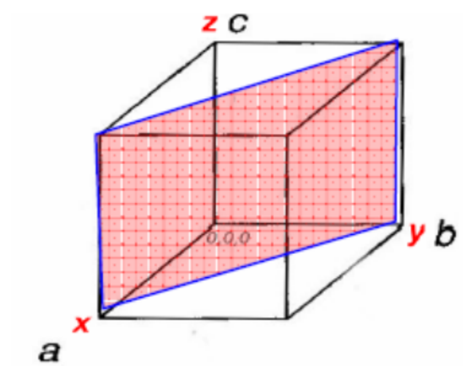
Face centered cubic (fcc)
has cubic symmetry.



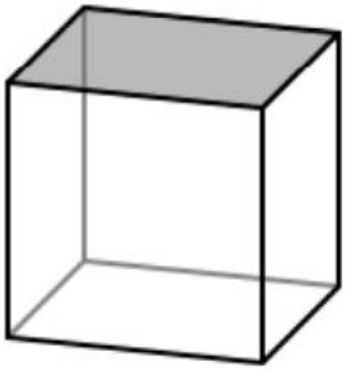
Face-centered cubic



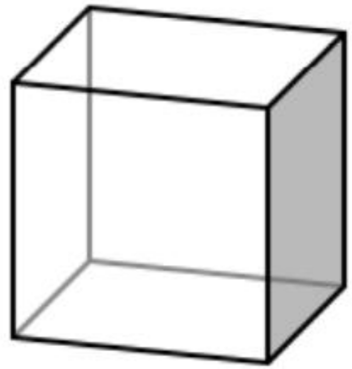
Miller Indices



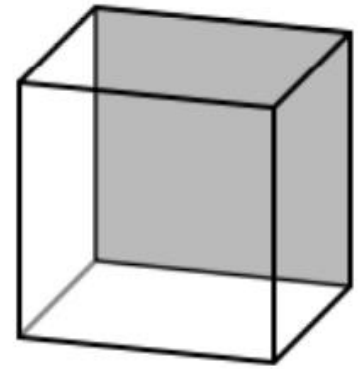
	x (a)	y (b)	z (c)
Intercept on axes	1	1	∞
Reciprocal	1/1	1/1	$1/\infty = 0$
Integer Clear	1	1	0
Miller Indices	(110)		



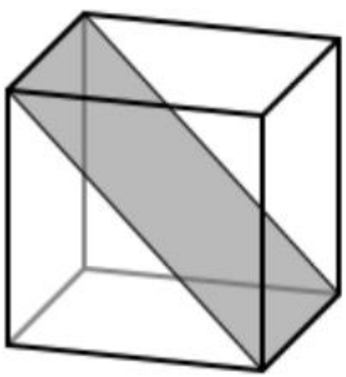
(001)



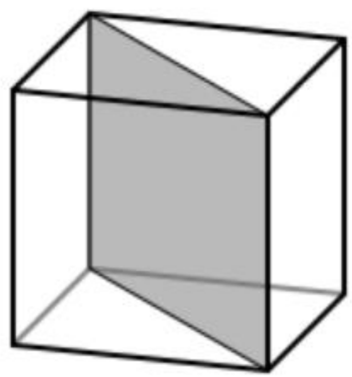
(100)



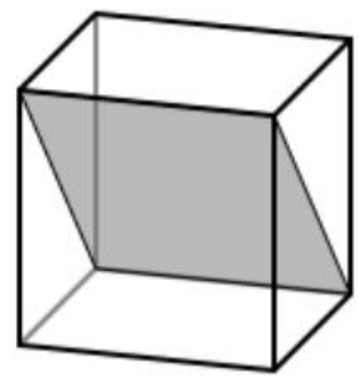
(010)



(101)



(110)



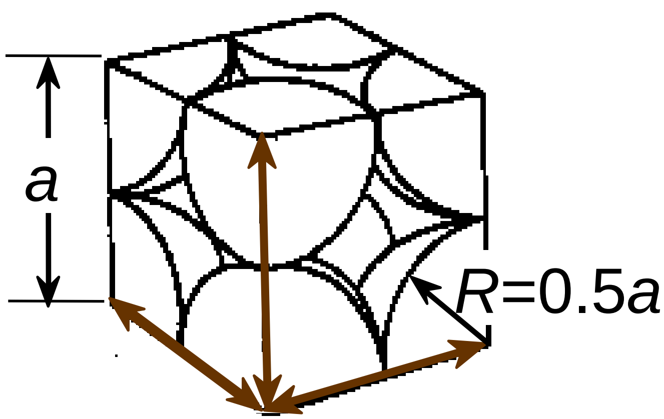
(011)

Atomic Packing Factor (APF)

$$\text{APF} = \frac{\text{Volume of atoms in unit cell*}}{\text{Volume of unit cell}}$$

*assume hard spheres

- APF for a simple cubic structure = 0.52



close-packed directions

contains 8 x 1/8 =

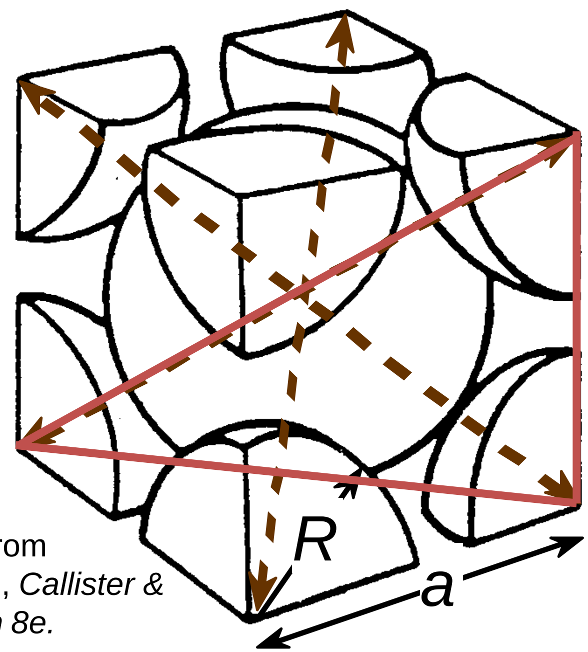
1 atom/unit cell

Adapted from Fig. 3.24,
Callister & Rethwisch 8e.

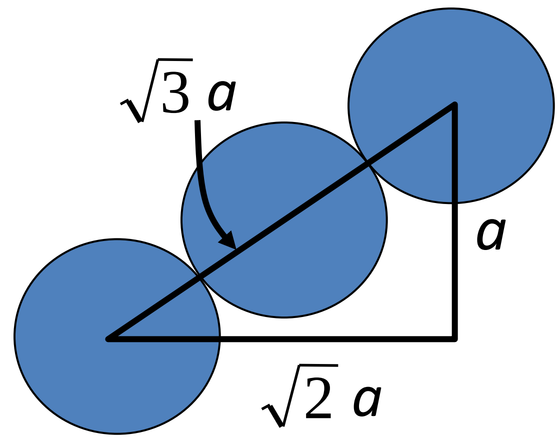
$$\text{APF} = \frac{\overbrace{1}^{\frac{\text{atoms}}{\text{unit cell}}}\ \overbrace{\frac{4}{3}\pi(0.5a)^3}^{\frac{\text{volume}}{\text{atom}}}}{\underbrace{a^3}_{\frac{\text{volume}}{\text{unit cell}}}}$$

Atomic Packing Factor: BCC

- APF for a body-centered cubic structure = 0.68



Adapted from Fig. 3.2(a), *Callister & Rethwisch 8e.*



Close-packed directions:
length = $4R = \sqrt{3}\,a$

atoms / unit cell

volume / atom

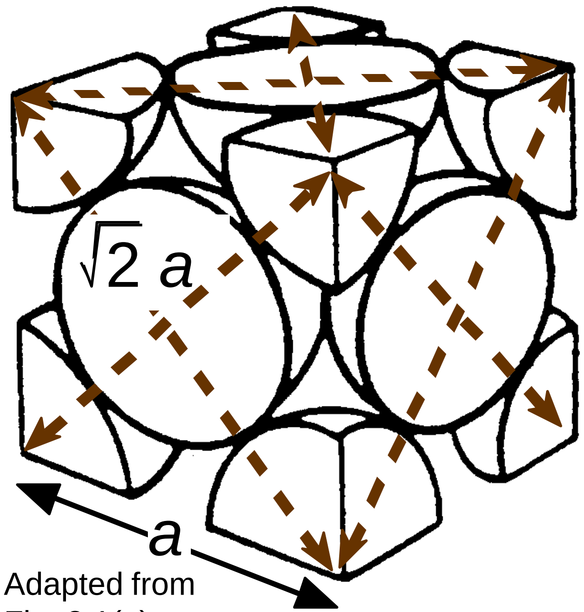
$$\mathrm{APF} = \frac{2\,\frac{4}{3}\pi\left(\sqrt{3}a/4\right)^3}{a^3}$$

volume / unit cell

Atomic Packing Factor: FCC

- APF for a face-centered cubic structure = 0.74

  maximum achievable APF



Adapted from Fig. 3.1(a), *Callister & Rethwisch 8e.*

Close-packed directions:

length = $4R = \sqrt{2}\,a$

Unit cell contains:

$6 \times 1/2 + 8 \times 1/8$

$= 4$ atoms/unit cell

$$\text{APF} = \frac{\overset{\text{atoms/unit cell}}{4}\ \ \overset{\text{volume/atom}}{\frac{4}{3}\pi(\sqrt{2}a/4)^3}}{\underset{\text{volume/unit cell}}{a^3}}$$

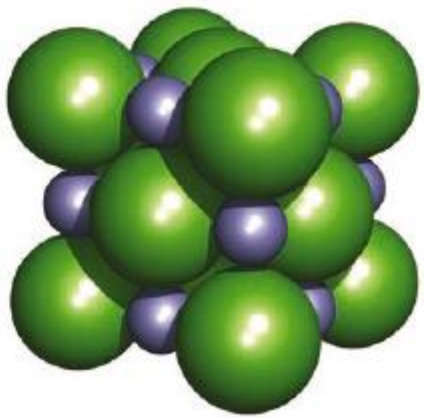
atoms / unit cell

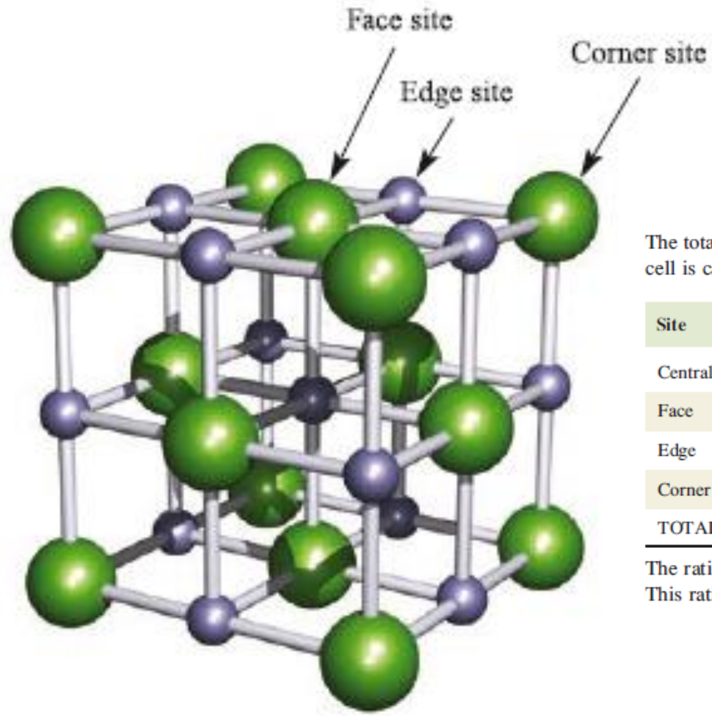
volume / atom

volume / unit cell

Simple Ionic Compounds

Rocksalt (NaCl) structure type





The total number of Na^+ and Cl^- ions belonging to the unit cell is calculated as follows:

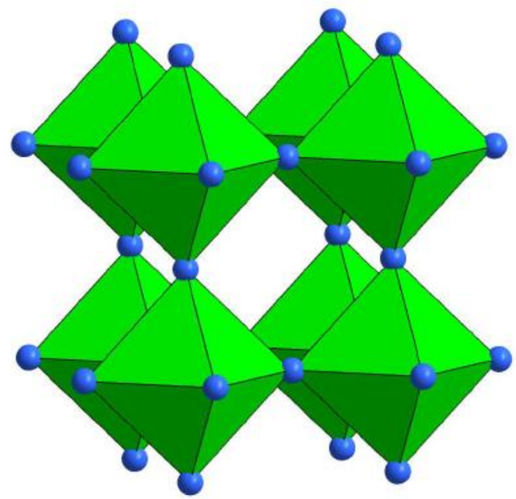
Site	Number of Na^+	Number of Cl^-
Central	1	0
Face	0	$(6 \times \frac{1}{2}) = 3$
Edge	$(12 \times \frac{1}{4}) = 3$	0
Corner	0	$(8 \times \frac{1}{8}) = 1$
TOTAL	4	4

The ratio of $Na^+ : Cl^-$ ions is $4:4 = 1:1$
This ratio is consistent with the formula NaCl.

Cl- Fcc arrangements , Na+ occupies octahedral holes

Among the many compounds that crystallize with the NaCl structure type are NaF, NaBr, NaI, NaH, halides of Li, K and Rb, CsF, AgF, AgCl, AgBr, MgO, CaO, SrO, BaO, MnO, CoO, NiO, MgS, CaS, SrS and BaS.

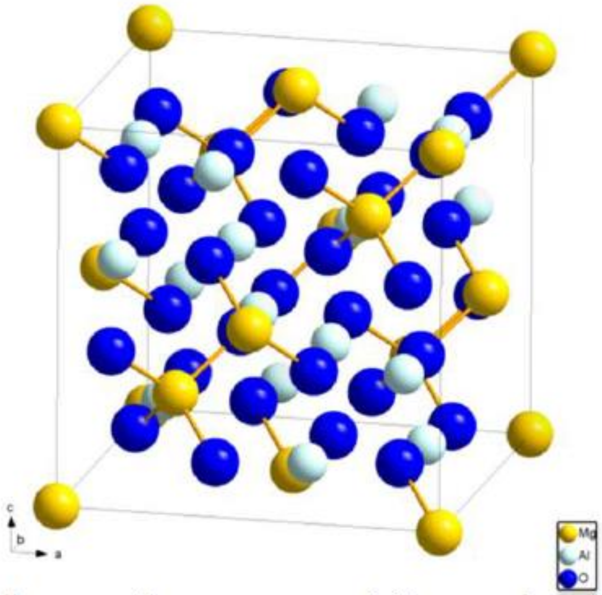
Structure Type – Rhenium Trioxide (ReO_3)

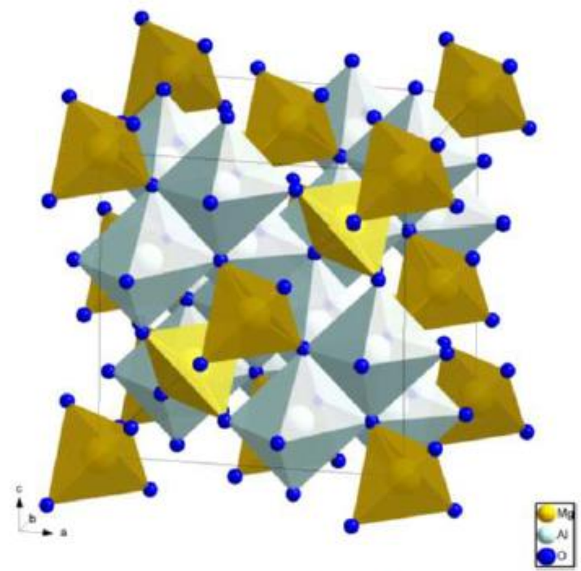


Also called the aluminum fluoride structure, which is adopted by the fluoride of Al, Sc, Fe, Co, Rh, and Pd, the high temperature form of WO_3, and ReO_3.

-Corner-sharing network of [ReO_6] that give a highly symmetrical structure with cubic symmetry.

Structure Type – Spinel ($MgAl_2O_4$)

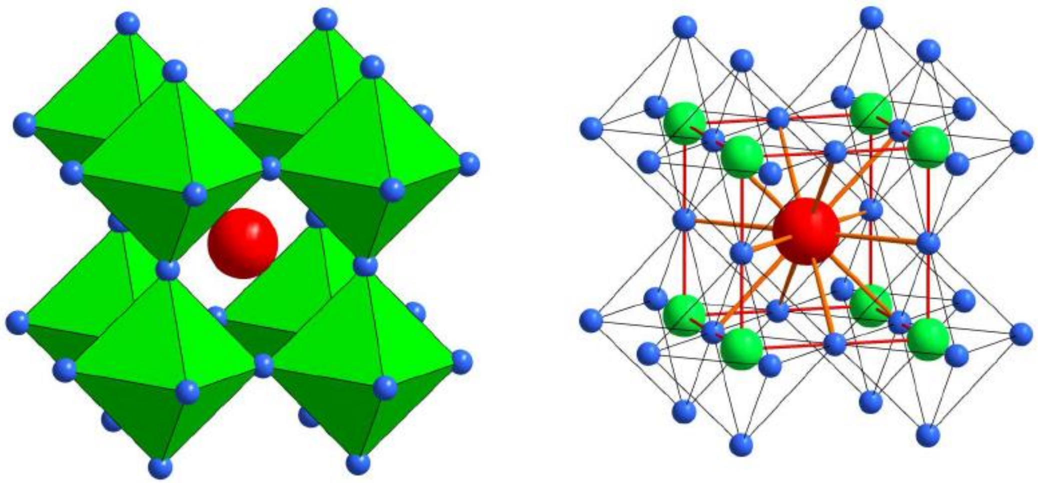




Spinels have the general formula AB_2O_4, where generally *A* is a divalent ion A^{2+}, *B* is trivalent ion B^{3+}.

The structure can be described as being based on a *cubic close-packed* array of oxide ions, with **A^{2+}** ions occupying tetrahedral holes and **B^{3+}** ions occupying octahedral holes.

Structure Type – Perovskite ($CaTiO_3$)



The structure is named after the mineral $CaTiO_3$.

-It has the same octahedral network as the ReO_3 structure, with the A ion added into the cavity.

-The structure may also be visualized as a *ccp* array of A and X ions with the B ions occupying the octahedral holes. (Compare with NaCl)

-Structures of high-temperature superconductors are based on perovskite.

Solid State Chemistry

Schottky defect

A *Schottky defect* is an example of a *point defect* in a crystal lattice and arises from vacant lattice sites. The stoichiometry and electrical neutrality of the compound are retained. Defects that fall in this category include:

- a vacant atom site in a metal lattice;
- a vacant cation and a vacant anion site in an MX lattice;
- two vacant cations and a vacant anion site in an M_2X lattice;
- a vacant cation and two vacant anion sites in an MX_2 lattice.

Crystal defects

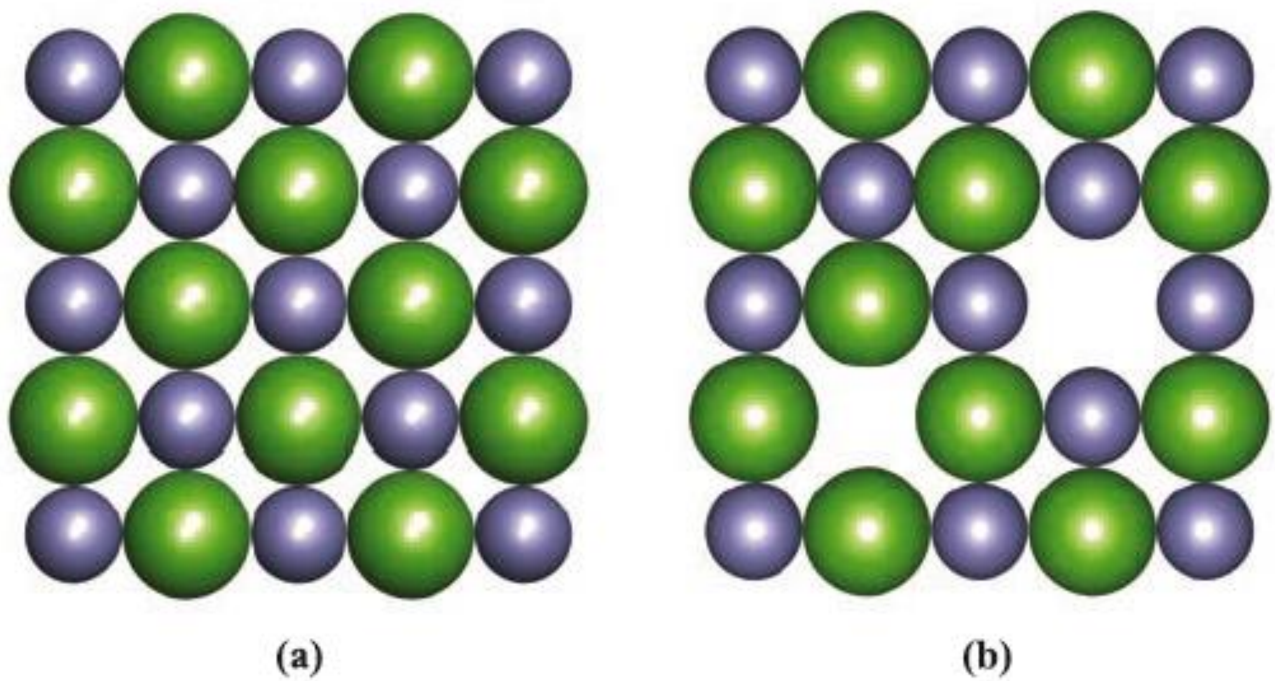


Fig. 6.27 (a) Part of one face of an ideal NaCl structure; compare this with Fig. 6.16. (b) A Schottky defect involves vacant cation and anion sites; equal numbers of cations and anions must be absent to maintain electrical neutrality. Colour code: Na, purple; Cl, green.

Crystal defects

Frenkel defect

In ionic lattices in which there is a significant difference in size between the cation and anion, the smaller ion may occupy a site that is vacant in the ideal lattice. This is a *Frenkel defect* (a *point defect*) and does not affect the stoichiometry or electrical neutrality of the compound.

Crystal defects

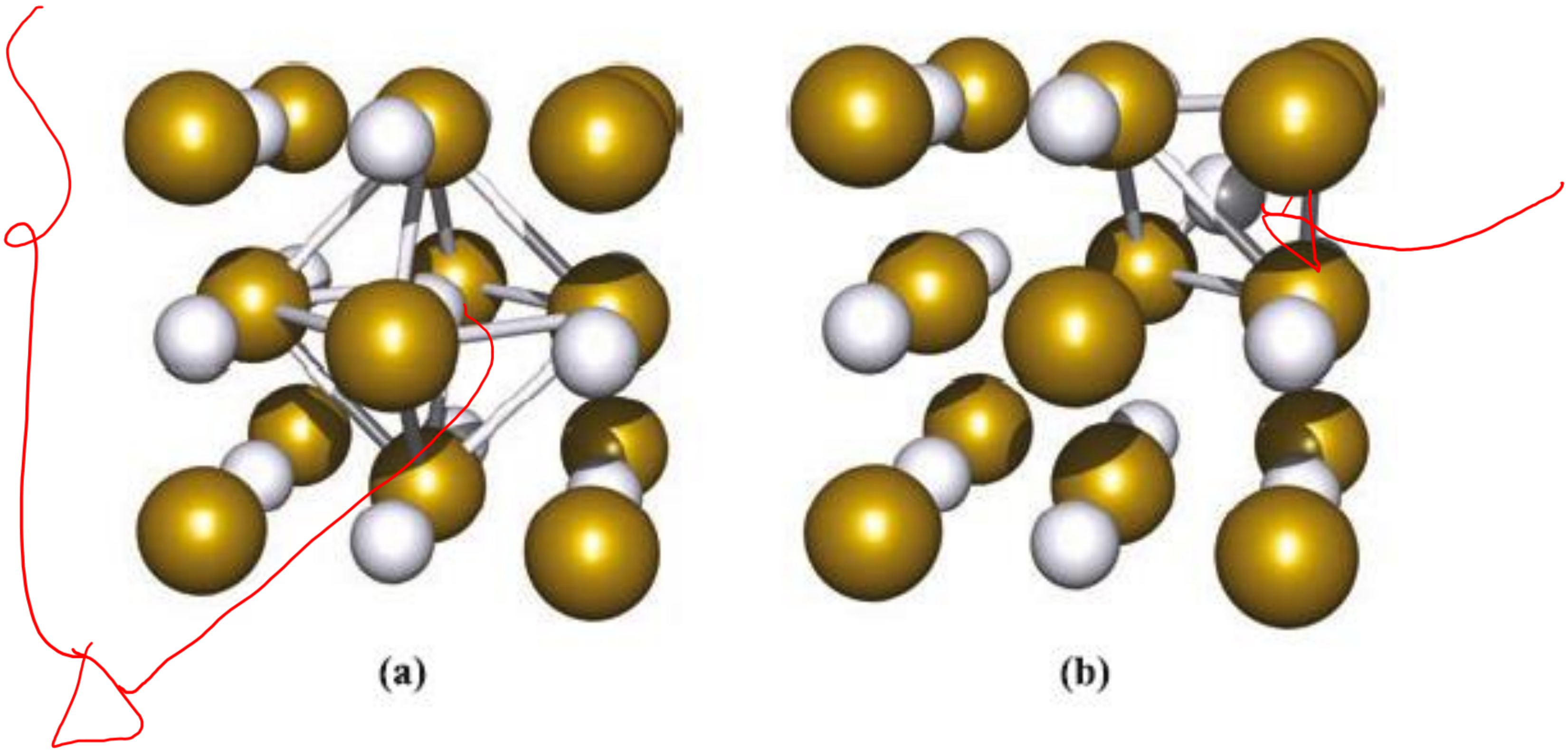
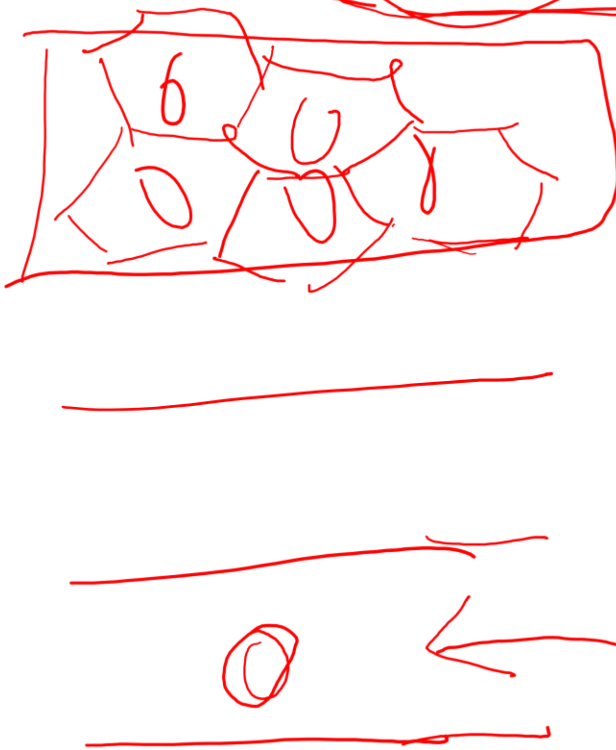


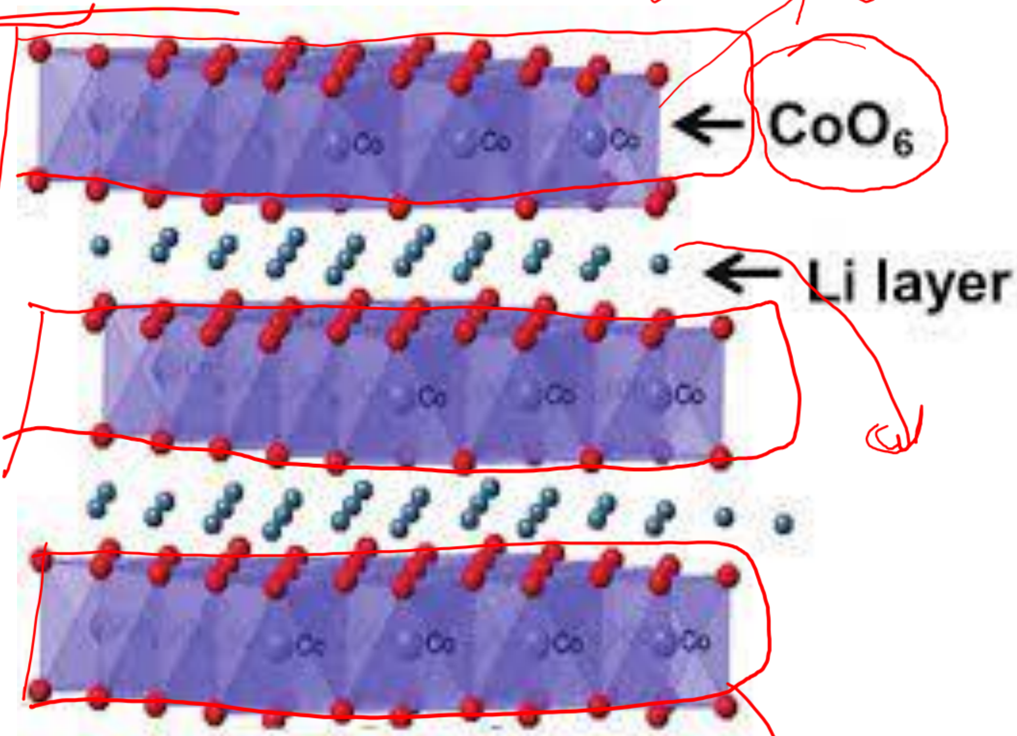
Fig. 6.28 Silver bromide adopts an NaCl structure. (a) An ideal lattice can be described in terms of Ag^+ ions occupying octahedral holes in a cubic close-packed array of bromide ions. (b) A Frenkel defect in AgBr involves the migration of Ag^+ ions into tetrahedral holes; in the diagram, one Ag^+ ion occupies a tetrahedral hole which was originally vacant in (a), leaving the central octahedral hole empty. Colour code: Ag, pale grey; Br, gold.

Structure – Property Relationship

$LiCoO_2$



Co



Graphite

Intercalation layer

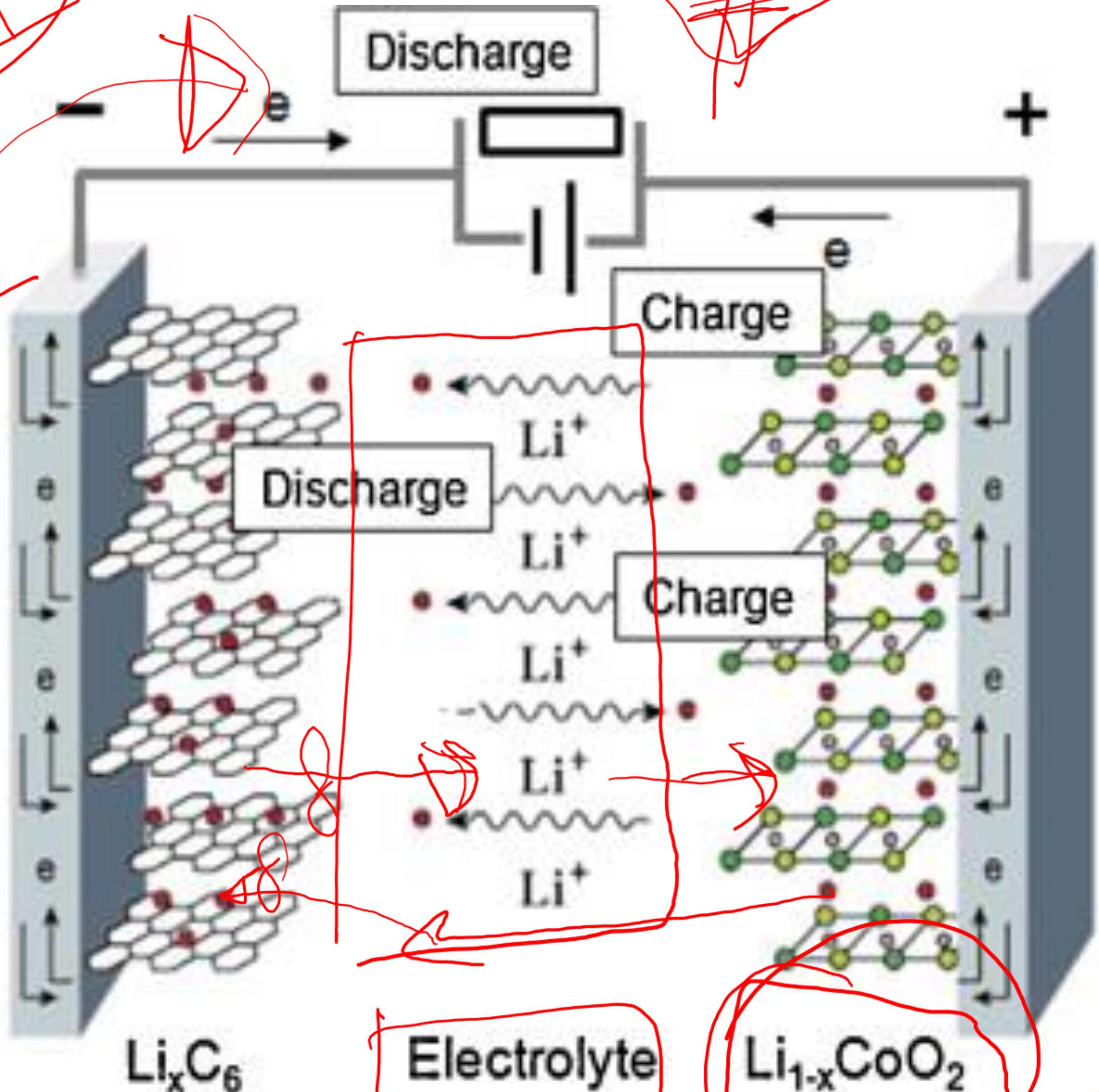
Secondary Li Ion Battery

Sony

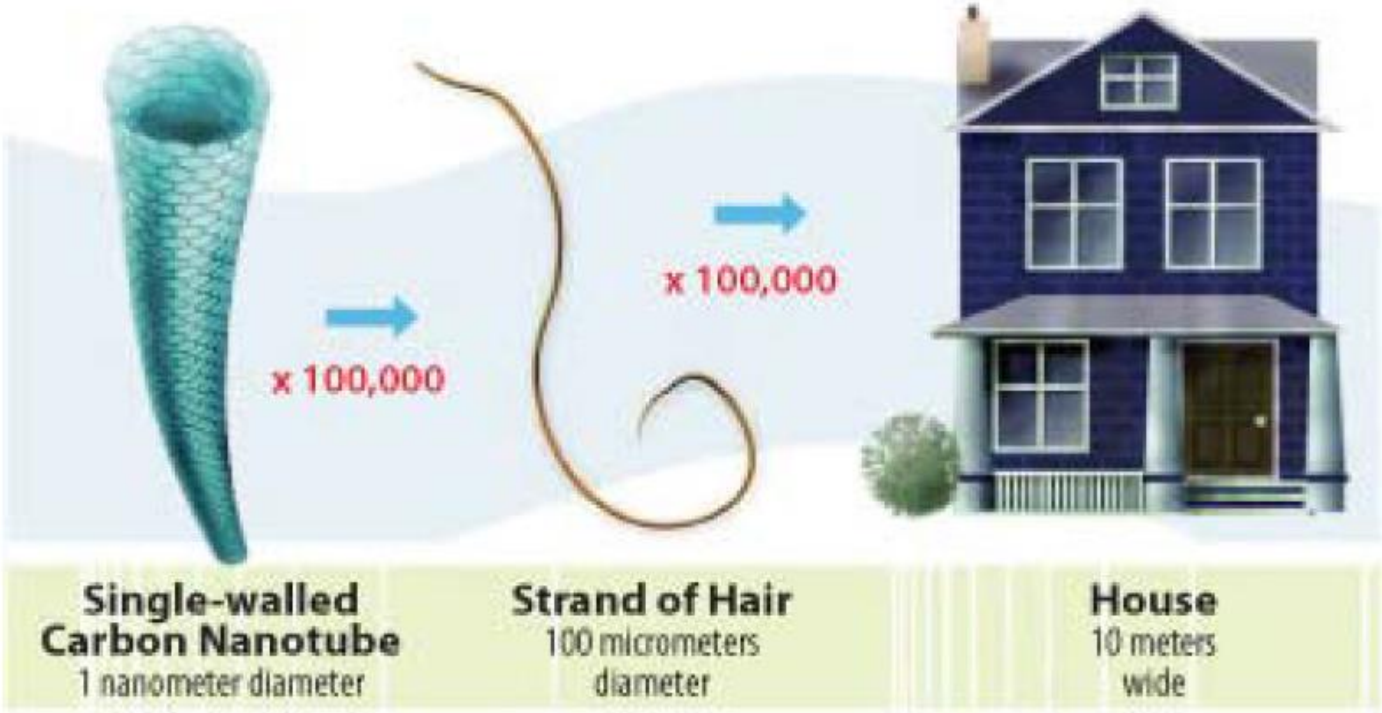
LiPF6

cycle

100%



Introduction to Nanomaterials


x 100,000
x 100,000
Single-walled Carbon Nanotube
1 nanometer diameter
Strand of Hair
100 micrometers diameter
House
10 meters wide

History of nanomaterials

SEM

- 1959: Richard Feynman' s famed talk." **There's Plenty of Room at the Bottom** "
- 1981: Binnig and Rohrer created the STM to image individual atoms. (Nobel, Physics 1986)
- 1985: Curl, Kroto, Smalley discovered **fullerene (Nobel, Physics 1996)**

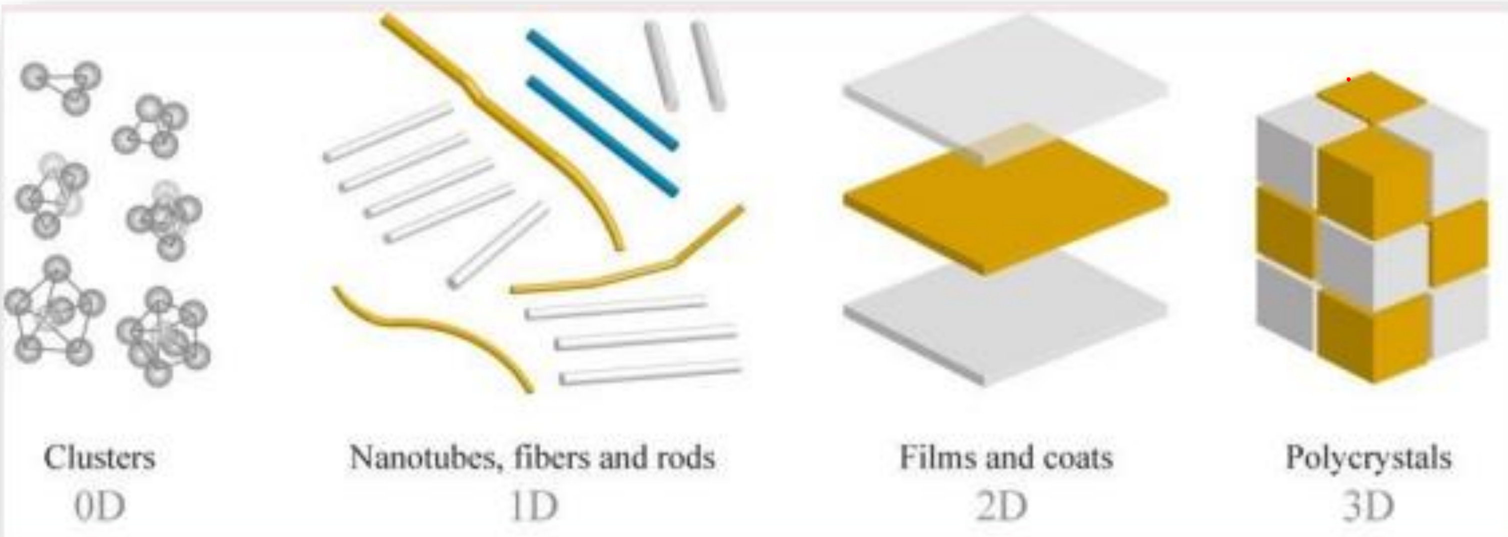
C60

TEM

- 1991: Iijima discovered single wall carbon nanotubes.
- 2010 A. Geim and K. Novoselov **(Nobel physics on Graphene)**

2016 Nobel Prize in Chemistry – development of **molecular machines** which can lead to more miniaturization and development of new materials.

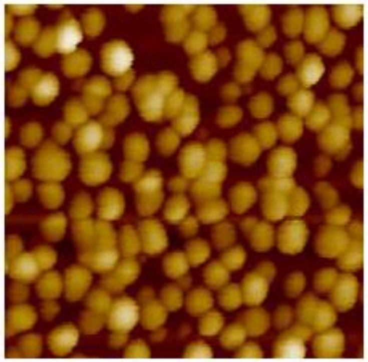
Classification of Nanomaterials: Based on Number of Free Dimensions



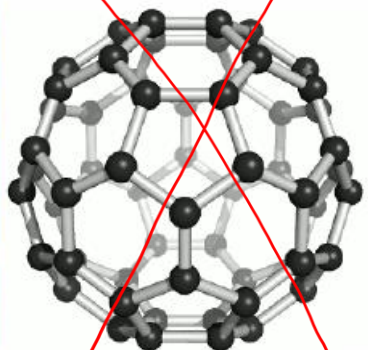
Classification of Nanomaterials: Based on Compositions

- Carbon-based nanomaterials
- Inorganic-based nanomaterials
- Organic-based nanomaterials
- Composite-based nanomaterials

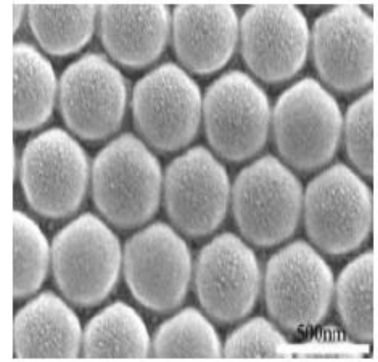
Nanomaterials (gold, carbon, metals, metal oxides and alloys) with variety of morphologies (shapes)



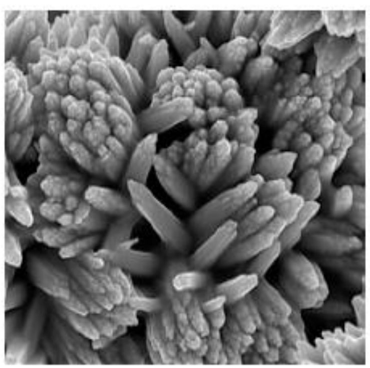
Au nanoparticle



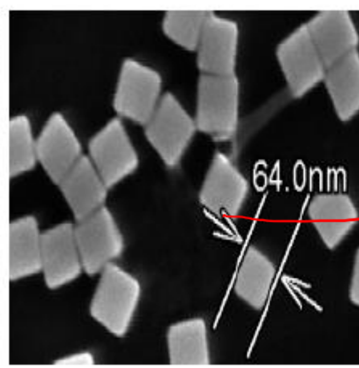
Buckminsterfullerene



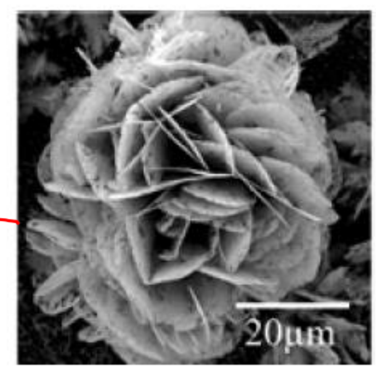
FePt nanosphere



Titanium nanoflower



Silver nanocubes

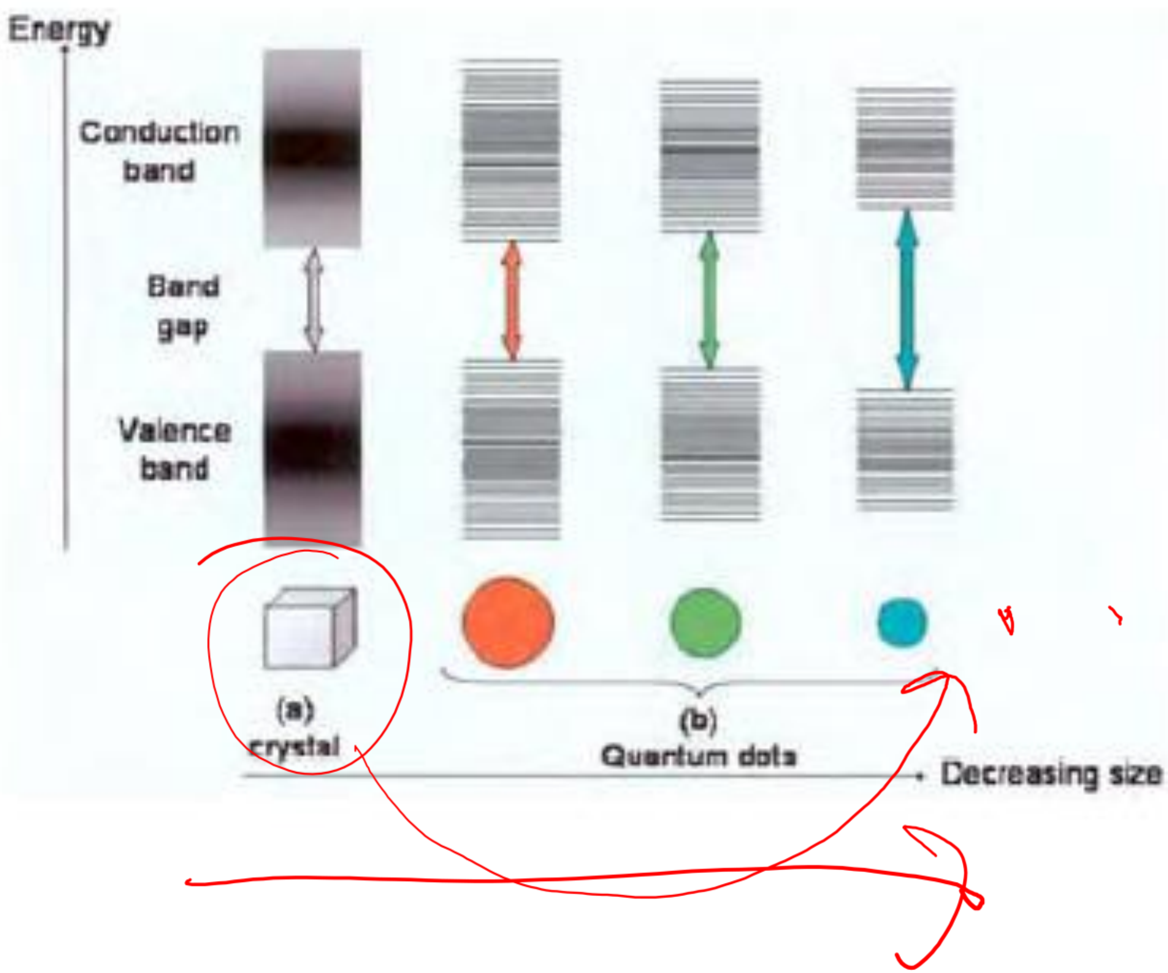


SnO2 nanoflower

Surface area increase

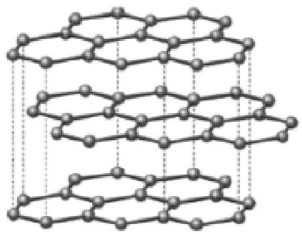
How would the total surface area increase if a cube of 1 m3 were progressively cut into smaller and smaller cubes, until it is composed of 1 nm3 cubes? Table (2) summarizes the results.

Size of cube side	Number of cubes	Collective surface area
1 m	1	6 m^2
0.1 m	1 000	60 m^2
0.01 m = 1 cm	10^6 = 1 million	600 m^2
0.001 m = 1 mm	10^9 = 1 billion	6 000 m^2
10^{-9} m = 1 nm	10^{27}	6 x 10^9 = 6 000 km^2


Energy
Conduction band
Band gap
Valence band
(a) crystal
(b) Quantum dots
Decreasing size

Carbon-based nanomaterials

Basic building block

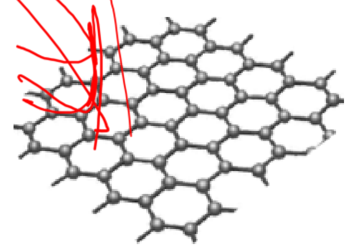


Graphite (3D)

van der Waals stack of graphene

Conductor

Stack



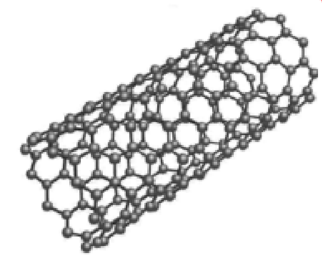
Graphene (2D)

Single-atom-thick carbon layer

sp2 bonding of carbons

Semi-metal

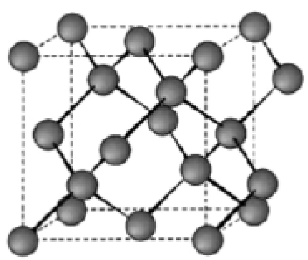
Roll



Carbon nanotube (1D)

Rolled graphene

Semiconductor (2/3) or metal (1/3)

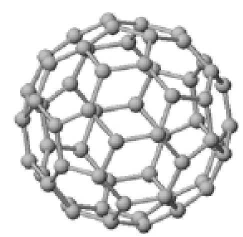


Diamond (3D)

sp3 bonding of carbons

Wide band gap (5.5 eV)

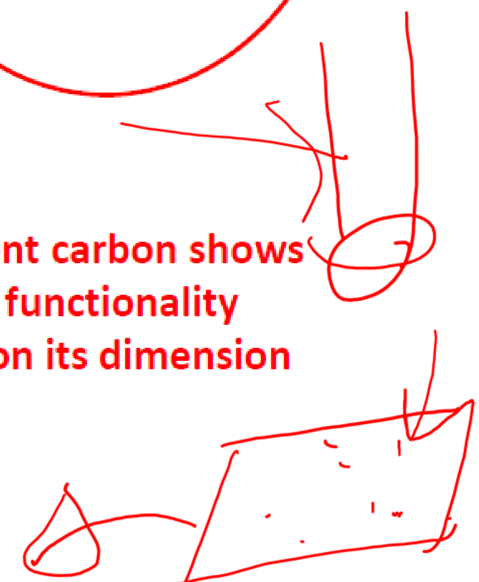
Ball



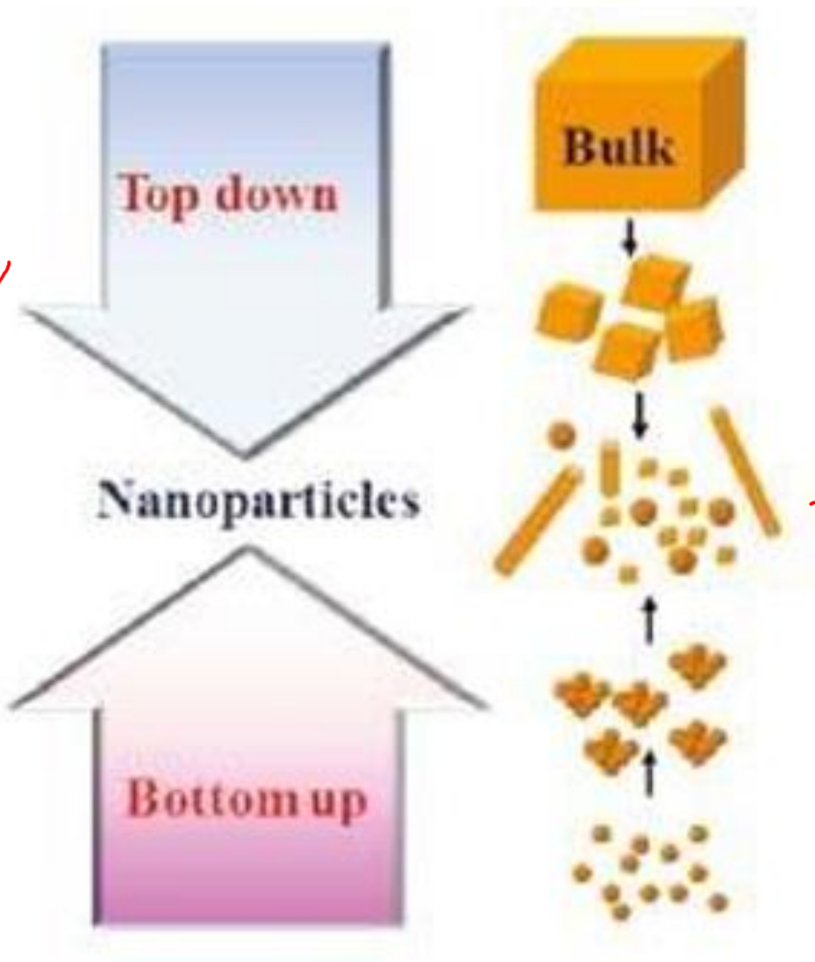
Fullerene (0D)

Wide band gap (5.5 eV)

Single element carbon shows different functionality depending on its dimension



Nanomaterial - synthesis and processing

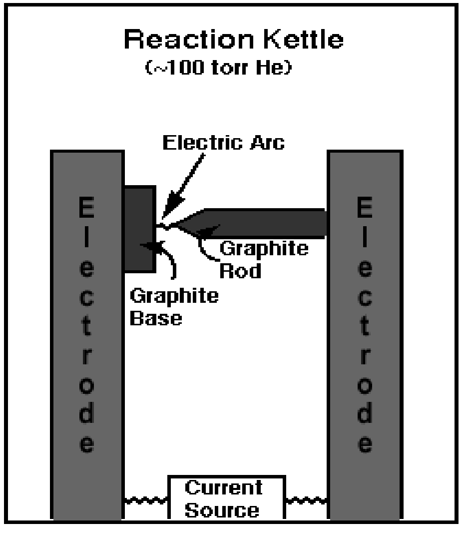


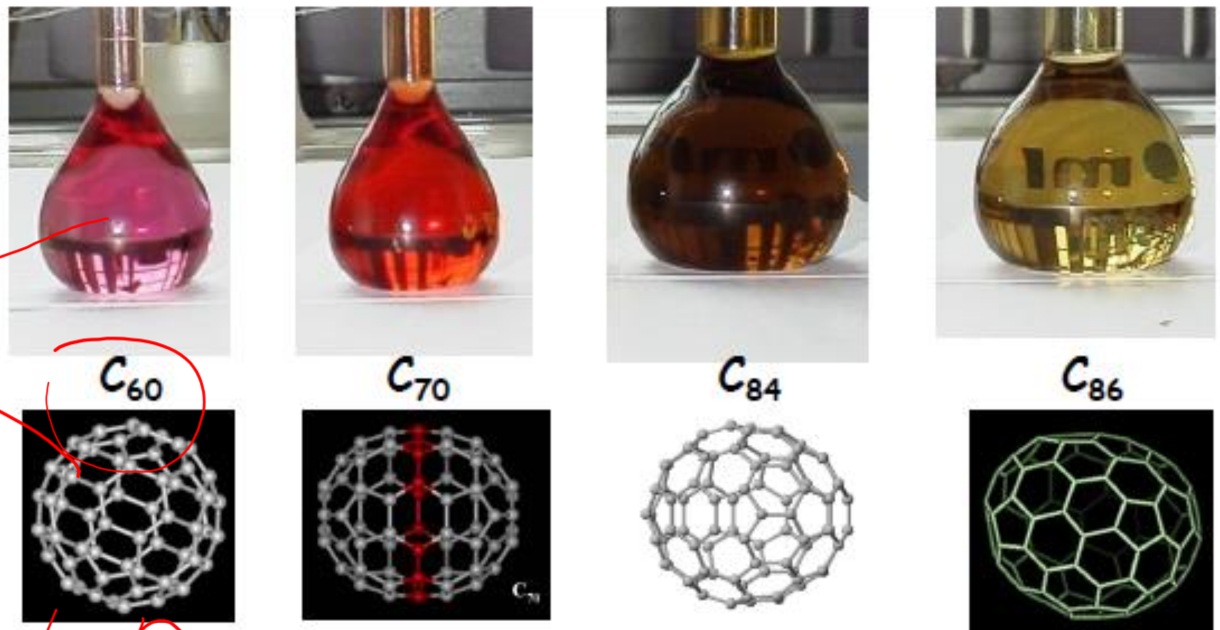
clusters

atoms

Fullerenes can be made by vaporizing carbon within a gas medium.

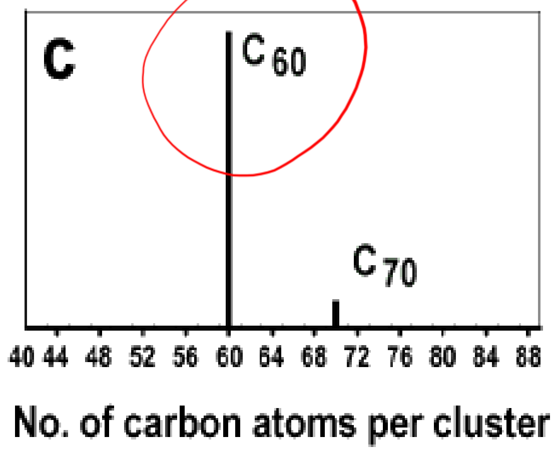
- An electric arc is maintained between two nearly contacting graphite electrodes.
- Most of the power is dissipated in the arc and not in resistive heating of the rod.
- The entire electrode assembly is enclosed in a reaction kettle that is filled with ~ 100 torr pressure of helium.
- Black soot is produced, and extraction with organic solvents yields fullerenes.



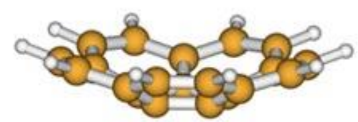


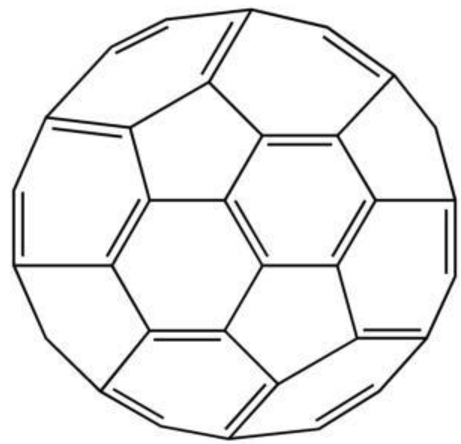
C_{60}

Structure C_{60}



corannulene

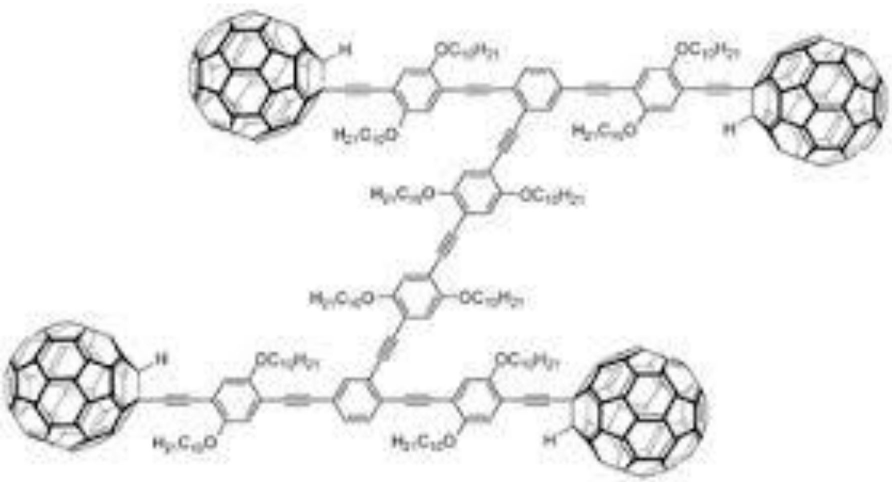


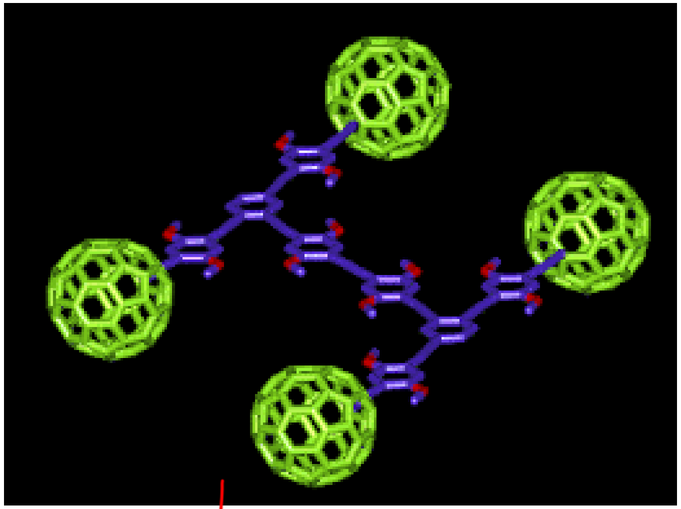


30 [6,6] double bonds connect two hexagons

60 [5,6] bonds connect a hexagon and a pentagon.

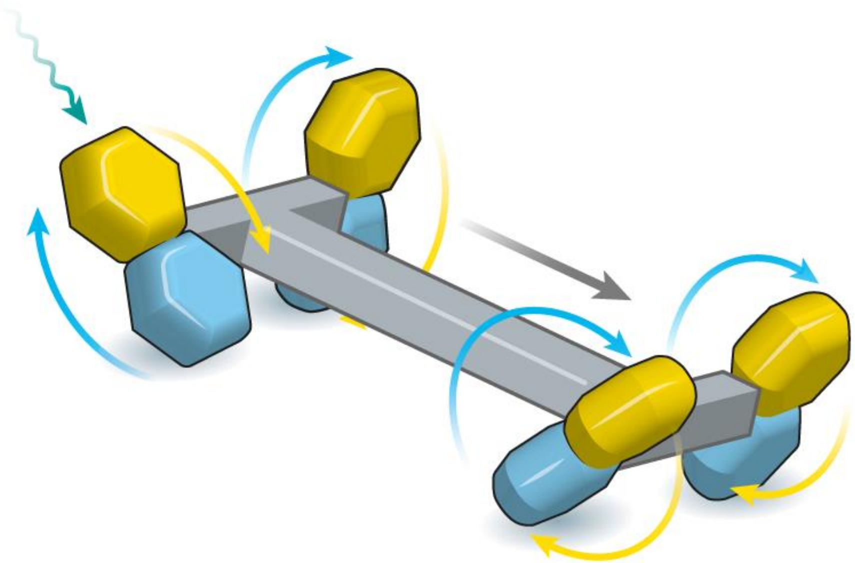
The X-ray diffraction bond length values are 135.5 pm for the [6,6] bond and 146.7 pm for the [5,6] bond





NOT FOR EXAM


"For the greatest benefit to mankind"
2016 NOBEL PRIZE IN CHEMISTRY
Jean-Pierre Sauvage
Sir J. Fraser Stoddart
Bernard L. Feringa



Nanotube

- Discovered 1991, Iijima